

8421

5 $\Rightarrow$ 0101
9 - 1001
A - 1010
:
FH - 1111

8bit

16bit

00H

0000H

FH - 1111

1
FFH

1
FFFFH

16 values

byte

word
8 bit Nos

00 $\Rightarrow$ 0000 0000

35H $\Rightarrow$ 0011 0101

41H $\Rightarrow$ 0100 0001

72 $\Rightarrow$ 0111 0010

FF $\Rightarrow$ 1111 1111

$$\begin{aligned}
 2^0 &= 1 \\
 2^1 &= 2 \\
 2^2 &= 4 \\
 2^3 &= 8 \\
 2^4 &= 16 \\
 2^5 &= 32 \\
 2^6 &= 64 \\
 2^7 &= 128 \\
 2^8 &= 256 \\
 2^9 &= 512 \\
 2^{10} &= 1K \\
 2^{20} &= 1M \\
 2^{30} &= 1G \\
 2^{40} &= 1T
 \end{aligned}$$

$$2^{11} \Rightarrow 2 \times 2^{10}$$

$$\Rightarrow 2 \times 1K$$

$$\Rightarrow \underline{2K}$$

$$2^{12} \Rightarrow 2 \times 2^{11}$$

$$\Rightarrow 4 \times 1K$$

$$\Rightarrow 4K$$

$$2^{16} \Rightarrow 2 \times 2^{15}$$

$$\Rightarrow 64 \times 1K$$

$$\Rightarrow \underline{64K}$$

$$2^{19} \Rightarrow 2 \times 2^{18}$$

$$\Rightarrow \underline{512K}$$



$$\begin{aligned} 2^{20} &= 2^{10} \times 2^{10} \\ &= 1K \times 1K \\ &= 1M \end{aligned}$$

0101
1001
1010

1111

05

000

101

001

010

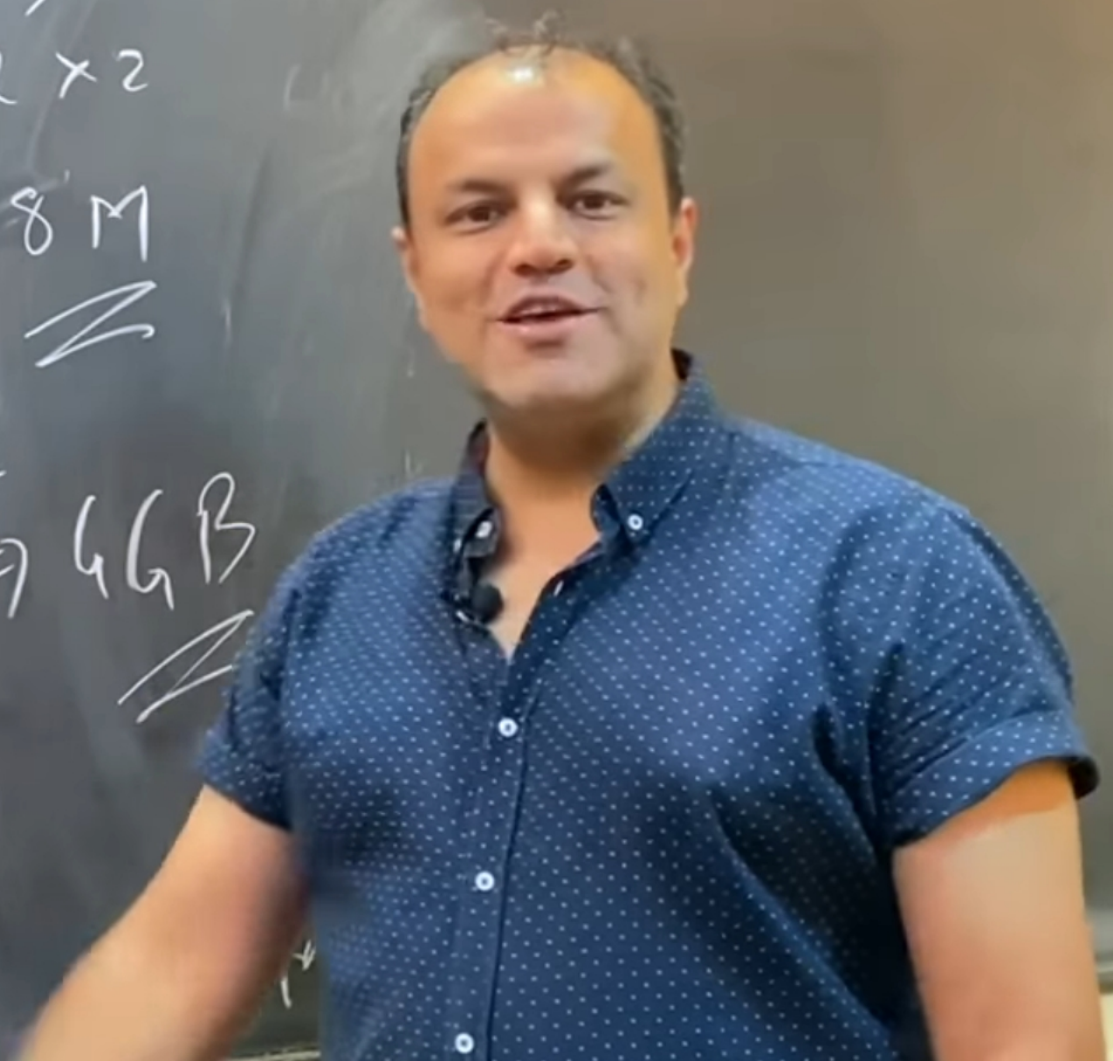
111

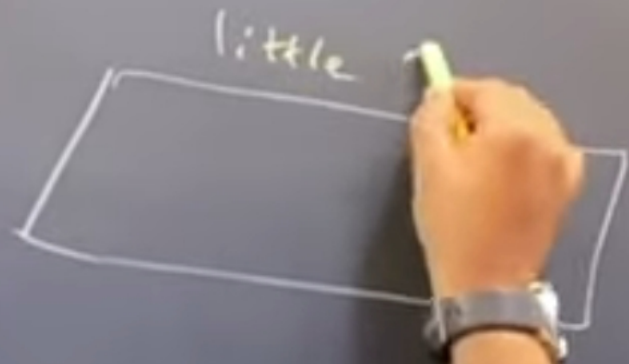
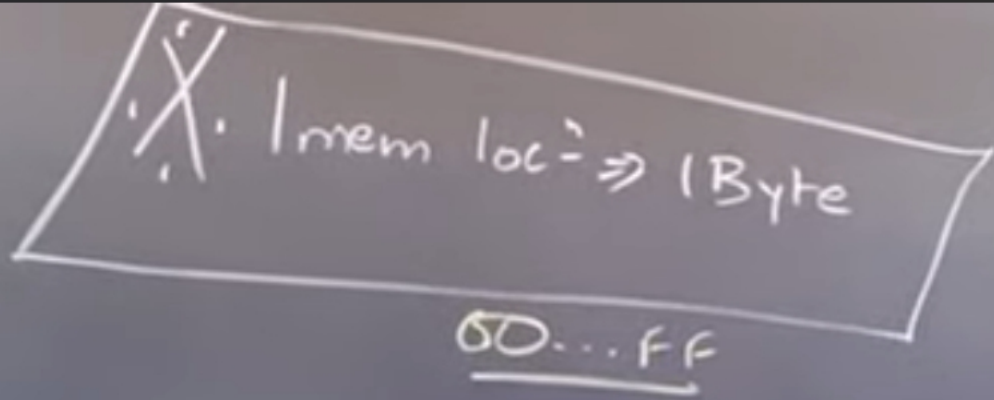
$$\left. \begin{array}{l} 2^4 = 16 \\ 2^5 = 32 \\ 2^6 = 64 \\ 2^7 = 128 \\ 2^8 = 256 \\ 2^9 = 512 \\ 2^{10} = 1K \\ 2^{20} = 1M \\ 2^{30} = 1G \\ 2^{40} = 1T \end{array} \right\}$$

$$2^3 \quad 3 \quad 2^0 \\ 2 \Rightarrow 2 \times 2$$

$$\Rightarrow 8M$$

$$32 \\ 2 \Rightarrow 64B$$

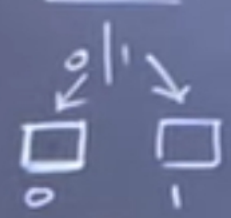




mem. accessed by 8085

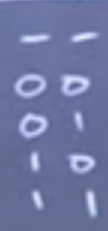
mem

1-bit



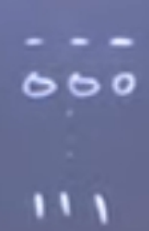
2 locations

2bit



4 loc^s

3bit

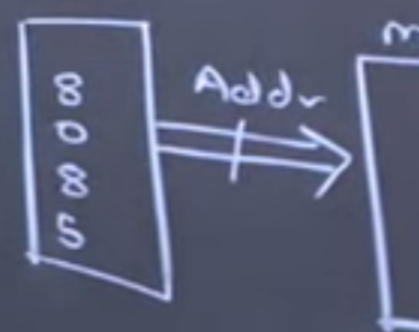


8 loc^s

n-bit add. bus

2ⁿ locations

2ⁿ Bytes of Mem



I/O

Addr. bus	mem
n-bits	2^n Bytes
6-bit	$2^6 = 16$ Bytes
6-bit	$2^6 = 64$ B
10-bit	$2^{10} = 1$ KB
16 bit	$2^{16} = 64$ KB
20 bit	$2^{20} = 1$ MB
32 bit	$2^{32} = 4$ GB
40 bit	$2^{40} = 1$ TB

⇒

8085 buses

→ Address bus: 16 bit

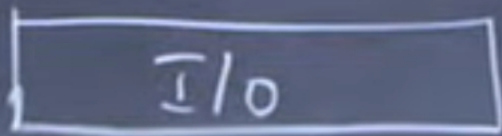
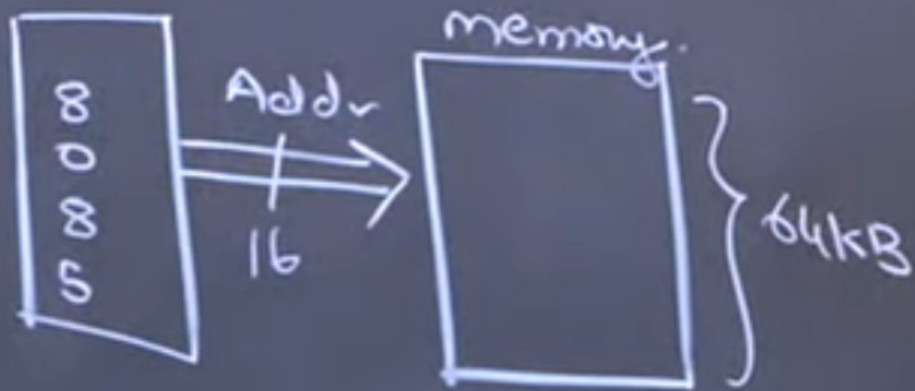
$$\therefore \text{mem} \Rightarrow 2^{16} = \underline{\underline{64KB}}$$

→ Data bus: 8 bit

ALU : 8 bit $\Rightarrow$ 8 bit up

→ Control bus: $\text{IO}/\overline{\text{M}}$, $\overline{\text{RD}}$, $\overline{\text{WR}}$





mem. accessed by 8085

64KB
first
last

Addr	Data
0000 H	
0001	
0002	
0003	
4000	
7124	
FFFF H	

min
= _____ H
max
= _____ H

1234

4000

512

mem. accessed by 8085

first
4KB
last

Addr	Data
0000 H	
0001	
0002	
0003	
4000	
7124	
FFFF H	

-16-1-8-

min
= 00 H

max
= FF H

1234

4000

5678



little Endian Rule

lower byte = lower
address

first

0000 H	34
0001	12
0002	00
0003	40
	18
4000	56
7124	32
FFFF H	

min
= 00 H

max
= FF H

HB | LB

12 | 34

4000

5678

32

last

-16-1-8-1

memory

$2^{16} = 64KB$